Jonghoon Ahn

3D Visualization & Animation Technical Artist

California | U.S. Permanent Resident / Work Authorized
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jonghoonahn.com/DigitalHuman_Portfolio_Jonghoonahn.html

Technical Artist specializing in digital humans, performance capture, and real-time character pipelines. Connects source preparation, likeness reconstruction, body and facial capture, motion cleanup, and final engine presentation. Works with artists and performers to resolve production problems across Maya, MotionBuilder, Unreal Engine, and Unity.

RELEVANT EXPERIENCE

Technical Artist / Digital Human Technical Director

Oct 2024 - Present

Yaloo | Contract

- Developed scan- and depth-based digital-human workflows across Maya, Wrap, MetaHuman, Unreal Engine, and Unity, including likeness preservation, texture transfer, eye alignment, hair integration, and character variants.
- For Digital Human Shininho (2025-2026), rebuilt an initial Unity feasibility character from the high-resolution scan after the prototype lost likeness; transferred the source to MetaHuman-compatible production geometry.
- Operated OptiTrack / Motive capture from marker placement and calibration through performer direction, live tracking review, retakes, and FBX delivery; supplemented weak wrist and finger motion with MOVIN TRACIN.
- Cleaned motion in MotionBuilder, reconstructed damaged segments with Maya HumanIK and animation layers, and assembled independently captured body / face takes in Unreal Sequencer.

Technical Artist

May 2026 - Aug 2026

Jung Hye-jeong | Row, Row, Row Your Boat | Contract

- Defined capture requirements for artist-recorded iPhone source footage and processed ten participants into individually validated MetaHuman identities; used supplemental Polycam head data when a Live Link source failed.
- Tuned skin, eyes, teeth, and grooms; assembled canon-driven character reveals and facial timing, corrected head / body alignment, and prepared the two-channel installation renders.

Technical Artist / Technical Director

Apr 2025 - Apr 2026

Erick Oh | Confidential Real-Time Game | Contract

- Developed Perlin-driven HLSL mesh deformation, C# directional-flow controls, and a raymarched metaball visual field with runtime contact / proximity logic in Unity URP.
- Authored four matching-topology cloud-growth states in Maya and integrated blendshape playback into Unity Timeline / Animation for predictable art direction.

CORE SKILLS

Digital humans
Maya, ZBrush, Wrap / ZWrap, MetaHuman, CC4, topology / UV transfer, facial blendshapes, rigging

Performance capture
OptiTrack / Motive, MOVIN TRACIN, iPhone / Live Link Face, MetaHuman Animator, Rokoko head rig

Animation / rendering
MotionBuilder, HumanIK, Control Rig, Sequencer, Unity Timeline, Unreal Engine, Unity HDRP / URP

Materials / development
XGen hair cards, Marvelous Designer, Alembic, skin / eye / groom lookdev, C#, HLSL, Python workflow integration

RELEVANT EXPERIENCE - CONTINUED

Lead Developer - Research Program
Oct 2019 - Sep 2022

Kookmin University | Museum / AR / MR Projects

- Led real-time AR / MR installation development for museum and research projects, connecting graphics, sensors, interaction, and exhibition deployment.
- For Gang Se-hwang (2022), reconstructed a portrait-based digital human with ZBrush / ZWrap, Maya UVs, and a custom facial library expanded from 52 ARKit base expressions to approximately 256 targets.
- Integrated and debugged an existing Python / Maya facial-data import workflow, authored XGen-to-card hair, delivered Marvelous Designer Hanbok simulation as Alembic, and built Unity HDRP lookdev and C# kiosk interaction.

ADDITIONAL CAPTURE, TEACHING & XR EXPERIENCE

Motion Capture Specialist
2026

When It Seen | Project

Integrated MOVIN TRACIN body motion and separately captured iPhone facial performance for three dancers in Unreal; delivered full-character Alembic geometry for downstream Houdini cloth simulation.

Technical Art Mentor / Workshop Instructor
Jan 2026 - Feb 2026

CultureHub LA

Led a month-long program with hands-on technical guidance, project feedback, and implementation support for artists working with real-time media.

Teaching Assistant - Graduate Program
Oct 2024 - May 2025

California Institute of the Arts

Supported MFA students with Unreal Engine 5, Unity AR, real-time rendering, debugging, and project production.

XR Game Developer
Jul 2024 - Sep 2024

RedFlight AI

Prototyped conversational-character scenes and XR interactions in Unity, connecting NPC dialogue behavior with real-time experience flow.

EDUCATION

PhD student, Media Arts and Technology — 2026 - Present
University of California, Santa Barbara

MFA, Art and Technology — 2023 - 2025
California Institute of the Arts

BFA, Entertainment Design — 2018 - 2022
Kookmin University